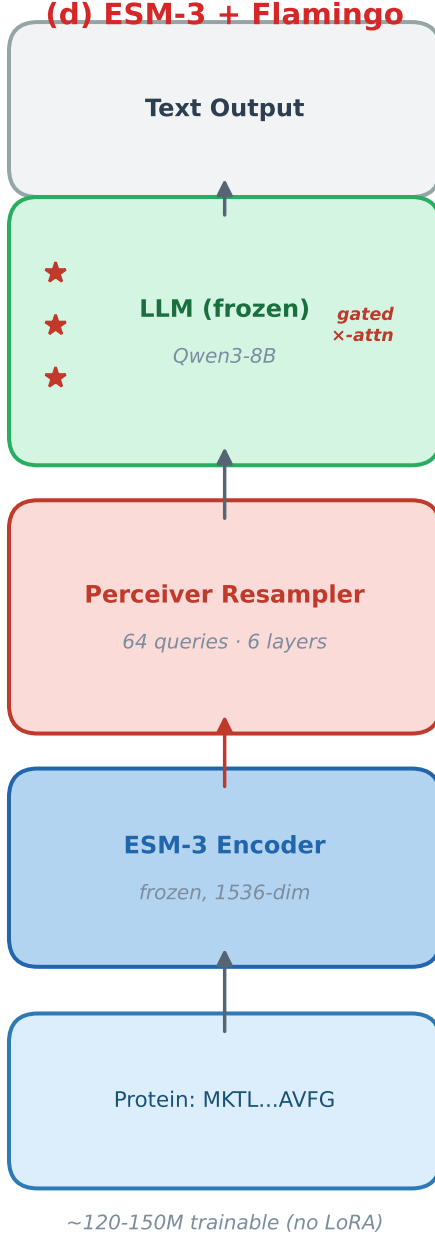
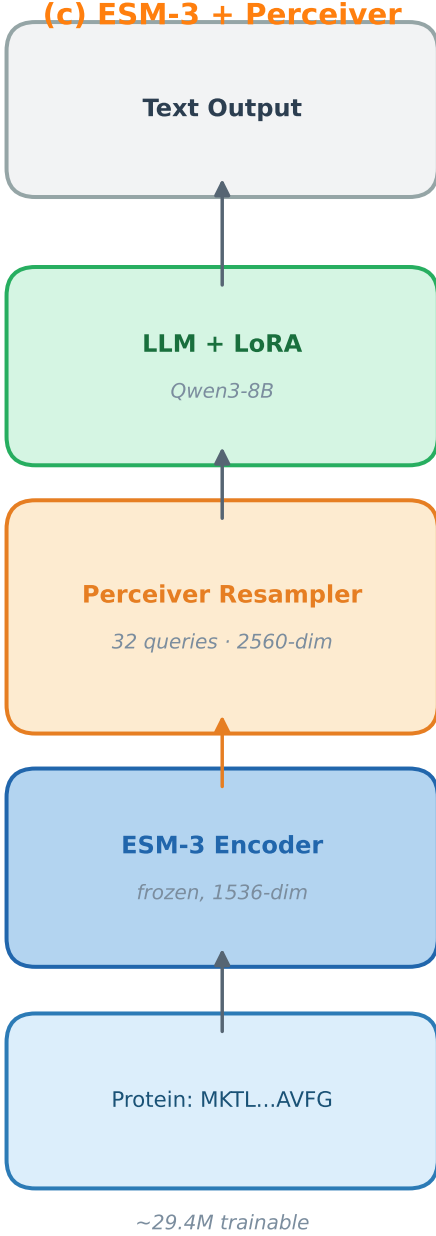
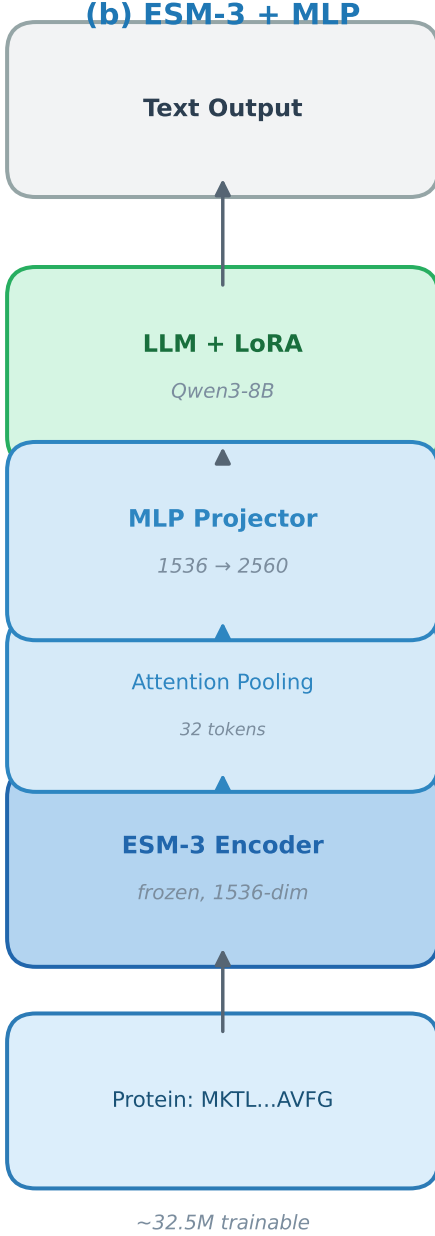
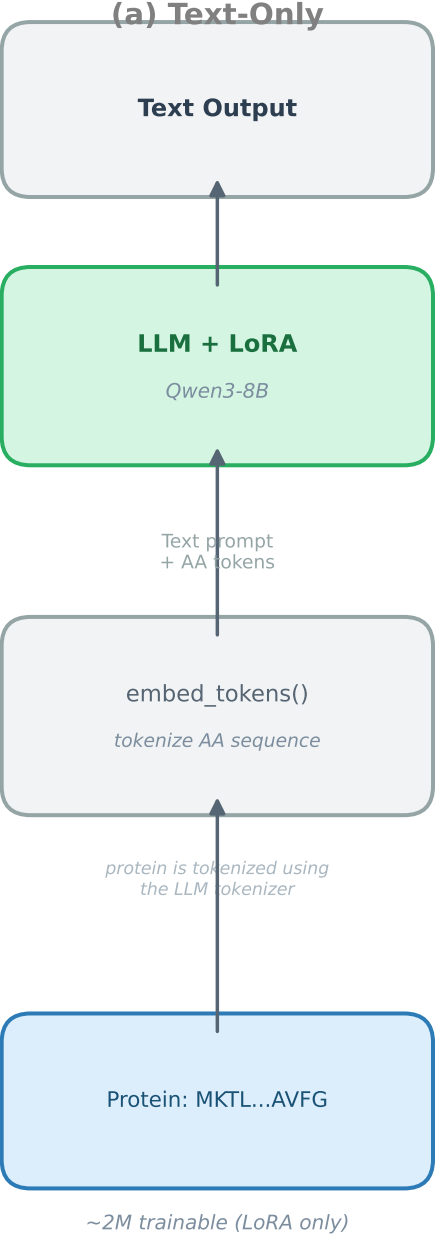


Protein-LLM Architecture: Four Approaches



Frozen (pretrained) Trainable LLM + LoRA Gated Cross-Attention